

Retrieving Contextual Information for Long-Form Question Answering using Weak Supervision

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Abstract

Long-form question answering (LFQA) aims at generating in-depth answers to end-user questions, providing relevant information beyond the direct answer. However, existing retrievers are typically optimized towards information that directly targets the question, missing out on such contextual information. Furthermore, there is a lack of training data for relevant context. To this end, we propose and compare different weak supervision techniques to optimize retrieval for contextual information. Experiments demonstrate improvements on the end-to-end QA performance on ASQA, a dataset for long-form question answering. Importantly, as more contextual information is retrieved, we find that the groundedness of generated long-form answers increases. Finally, we show that long-form answers often anticipate likely follow-up questions, via experiments on a conversational QA dataset.

1 Introduction

The goal of long-form question answering (LFQA) is to provide in-depth answers to end-user questions (Stelmakh et al., 2022; Fan et al., 2019). For example, for the user question

“When did Lionel Messi start his career?”

a direct answer would be:

16 November 2003

which is the date of his first team debut.

However, this answer naturally sparks a series of follow-up questions to obtain more contextual details (Kumar and Joshi, 2017):

“For which club?”

“In which match?”

“What about his La Liga debut?”

“How did his career develop?”

A long-form answer aims to proactively supply a more complete and detailed response beyond the

succinct direct answer, essentially anticipating such follow-up questions:

Lionel Messi started his career as a professional football player with FC Barcelona. He made his first-team debut in a friendly against Porto on 16 November 2003, at the age of 16. ...

We thus draw parallels between LFQA and conversational question answering (ConvQA) (Vakulenko et al., 2021; Voskarides et al., 2020; Qu et al., 2020; Coman et al., 2023) and hypothesize that they can be treated as complementary tasks.

Existing approaches for LFQA based on retrieval-augmented generation (Lewis et al., 2020; Izacard and Grave, 2021; Guu et al., 2020) typically utilize retrievers that are optimized to obtain direct answers to questions. Such retrieval systems thus often fail to retrieve relevant context, which is needed to generate faithful and comprehensive long-form answers.

Approach. We propose to train a specialised retriever for the task of LFQA that not only retrieves direct answers for a question, but also retrieves additional context required for grounding long-form answers. Our goal is to retrieve both direct answers and contextual information in one shot.

The major bottleneck for training a retriever for LFQA is the absence of training data. LFQA datasets (Fan et al., 2019; Stelmakh et al., 2022) contain questions and long-form answers but not the ground-truth passages required to produce those long-form answers. In this work, we propose a mechanism to automatically infer *silver passages*, designed to ground both (i) the direct answers, **and** (ii) the contextual information. These passages should provide sufficient evidence to support information in the long-form answer. This is different from previous work on factoid short-form QA, which identified such passages by matching only against the direct answers (Shen et al., 2023).

Based on such silver passages, we train BERT-

based re-ranking models (Nogueira and Cho, 2019). These re-rankers are applied to the initial retrieval results, to enhance recall of contextual information in the top-ranked passages. These top-ranked passages are then provided as input to an LLM to generate the long-form answer.

We conduct experiments on ASQA (Stelmakh et al., 2022), a dataset for LFQA, and show that we substantially improve end-to-end QA performance, while also increasing the groundedness of the long-form answer w.r.t. the retrieved passages.

Experiments on the ConvQA dataset CONVMIX (Christmann et al., 2022) demonstrate that our method generates long-form answers that often also contain answers to the follow-up questions, when provided only with the first question of the conversation as input. This indicates that LFQA can indeed anticipate likely follow-up questions.

Contributions.

- a novel mechanism using the target long-form answers to identify silver passages expressing relevant contextual information.
- improving end-to-end QA performance, achieving state-of-the-art performance on the competitive ASQA benchmark.
- an investigation into the relationship between the LFQA and ConvQA tasks as alternatives for satisfying the same information needs.

2 Identifying silver passages

The key idea for obtaining such silver passages is to utilize both the long-form answers (LFAs) and direct answers (DAs), as annotated in the ASQA dataset, jointly as a weak supervision signal for passage relevance. We retrieve a large set of passages first (say 100), using first-stage retrieval (Karpukhin et al., 2020), and then choose up to k silver passages from this candidate pool.

We considered three techniques (see Appendix A for details) for matching candidate passages against an LFA: (i) lexical matching, (ii) semantic similarity, and (iii) LLM perplexity. Our proposed approach matches against a combination of both LFAs and DAs, and we also compare these against matching only with DAs.

Lexical matching with LFA. We compute the matching score for a pair of a candidate passage p and the LFA as:

$$match(p, LFA) = \frac{|tokens(p) \cap tokens(LFA)|}{|tokens(LFA)|} \quad (1)$$

where *tokens* are sets of words produced by a tokenizer with stopwords removal.

Semantic similarity with LFA. We use a pre-trained Sentence-Transformer model¹ (Reimers and Gurevych, 2019; Vaswani et al., 2017), to compute the semantic similarity.

LLM perplexity of LFA. Inspired by the approach used in Toolformer (Schick et al., 2023), we compute the LLM perplexity of the target LFA (of length n) with the different candidate passages provided as context, as score for the candidate passage.

Matching with DA. All candidate passages that contain one of the DAs are considered as relevant. Here we consider exact lexical matches since the DAs are relatively short. This is the typical approach when the goal is to provide crisp answers (Shen et al., 2023).

SILVER: Matching with LFA & DA. Finally, we consider the combination of matching both the LFA and the set of DAs, for selecting k silver passages. First, we ensure that each DA is matched by at least one of the candidate passages. In case there are multiple candidate passages matching the same DA, the one with the highest matching score with the LFA is chosen as relevant. The remaining passages are chosen based on their LFA matching score, to obtain a total of k silver passages. We utilize lexical matching with the LFA, which showed strong results (see Sec. 3) and is computationally inexpensive. This combined variant is the SILVER approach proposed in this work, as it jointly optimizes towards information for directly answering the question and relevant contextual information.

3 Experiments

Benchmarks. We conduct experiments on two datasets: (i) ASQA (Stelmakh et al., 2022), a dataset for LFQA, and (ii) CONVMIX (Christmann et al., 2022), a dataset for ConvQA. We use ASQA for fine-tuning the re-rankers and LLMs.

Retrieval metrics. For evaluating retrieval, we measure recall in the top-5 retrieved passages. On ASQA, we compute *Direct Answer Recall* as the fraction of DAs appearing in the retrieved passages. As a proxy for recall of contextual information, we measure *Wikipedia Recall* as the fraction of the ground-truth relevant Wikipages matched with our retrieval results. A Wikipedia is matched if we retrieve a passage from the respective page. On

¹<https://huggingface.co/sentence-transformers/nli-roberta-base-v2>

Metric → Retrieval Method ↓	ASQA (Stelmakh et al., 2022)						CONVMIX (Christmann et al., 2022)			
	Recall: Direct Ans.	Recall: Wikipedia	Ground.	ROUGE-L	D-F1	DR	Recall: Direct Ans.	Recall: Follow-up Ans.	Ground.	C-F1
DPR (Karpukhin et al., 2020)	0.489	0.450	0.763	42.5	27.7	34.3	0.558	0.260	0.726	20.2
Matching with DA	0.498	0.483	0.810	42.9	28.0	34.7	0.622	0.292	0.803	20.8
Lexical matching with LFA	0.489	0.501	0.851	43.3	28.3	35.0	0.635	0.306	0.828	20.8
Semantic similarity with LFA	0.482	0.500	0.829	42.8	27.2	34.1	0.619	0.303	0.817	20.4
LLM perplexity of LFA	0.483	0.495	0.845	43.3	26.7	34.0	0.625	0.305	0.838	20.7
SILVER: Matching with LFA & DA	0.491	0.516	0.858	43.4	29.0	35.5	0.635	0.306	0.839	20.8

Table 1: Main results comparing retrieval and end-to-end QA performance on ASQA and CONVMIX.

Method	Length	ROUGE-L	D-F1	DR
FLARE (Jiang et al., 2023)	—	34.3	28.2	31.1
PaLM 540B (Amplayo et al., 2023)	64.1	40.7	27.8	33.5
JPR + T5-large (Stelmakh et al., 2022)	71.6	43.0	26.4	33.7
SIXPAQ (Sun et al., 2023)	63.5	43.8	28.9	35.6
DPR + VICUNA 13B	70.5	43	28.3	34.9
SILVER + VICUNA 13B (proposed)	70.0	44.1	30.8	36.9

Table 2: Comparison with state-of-the-art on ASQA.

CONVMIX, we measure *Direct Answer Recall* / *Follow-up Answer Recall* as the fraction of DAs for the first question / follow-up questions appearing in the retrieved passages.

Metrics. On ASQA, we keep the metrics used in the original work. This includes *ROUGE-L* (Lin, 2004) for measuring the overlap of the generated text with one of the two reference LFAs. For answer correctness, the Disambig-F1 (*D-F1*) metric is used, measuring the fraction of question interpretations answerable from the generated LFA. A pre-trained machine reading comprehension (MRC) (Hermann et al., 2015) model is used to this end. The *DR* metric combines the two metrics via the geometric mean.

For CONVMIX, inspired by the *D-F1* metric, we compute *C-F1* as the fraction of conversational questions answerable from the generated LFA. We use the same MRC model as for the *D-F1* metric.

In addition, we measure *Groundedness* as the fraction of tokens in the answer that is also present in the retrieved passages. Similar to lexical matching in Sec. 2, stopwords are not considered.

Configuration. We use DPR (Karpukhin et al., 2020)² for first-stage retrieval, which was shown to outperform BM25 (Robertson and Zaragoza, 2009) on ASQA in previous work (Sun et al., 2023).

We use Vicuna 13B 1.5 (Zheng et al., 2023) as LLM for generating LFQAs and fine-tune it for 1 epoch using the long-form answers in the ASQA dataset as the target output. The input to the model are the question and the top-5 passages retrieved either by DPR or by our SILVER re-rankers. We

²DPR is trained on NQ (Kwiatkowski et al., 2019) and used along the corresponding Wikipedia dump.

use the same prompt as in the original ASQA paper to combine the question and the retrieved passages.

Further details on the datasets and experimental setup in Appendix B.

3.1 Results

Recall of contextual information is enhanced. Table 1 shows our main results. The foremost take-away is that recall of contextual information is greatly improved when incorporating our SILVER re-rankers, compared to DPR. On ASQA, Wikipedia recall increases from 0.450 to 0.516. On CONVMIX, follow-up answer recall is improved from 0.260 to 0.306, indicating that our SILVER re-rankers aid the LLM to anticipate and successfully answer follow-up questions.

Lexical matching shows strong performance. An interesting finding is that simple lexical matching achieves better performance (DR of 35.0) compared to the more complex and computationally expensive variants based on semantic similarity (DR of 34.1) or LLM perplexity (DR of 34.0). This result demonstrates that lexical matching is sufficiently robust for the relatively long answers.

Improving end-to-end QA performance. Combining LFA and DA to provide supervision signal for training the re-ranker leads to the best performance (DR of 35.5), substantially improving over the baseline results with DPR (DR of 34.3). Both improvements in answer formulation (ROUGE-L of 43.4 vs. 42.5) and provision of the right answer in an appropriate context (D-F1 of 29.0 vs. 27.7) contribute to this overall increase in performance.

Groundedness is substantially enhanced. Another key take-away is the effect of our SILVER re-rankers on the groundedness of the generated answers. The groundedness is dramatically improved on both datasets compared to DPR retrieval (0.763 to 0.858 on ASQA and 0.726 to 0.839 on CONVMIX). This result indicates that our generated LFAs are more likely to be based on the retrieved passages rather than hallucinated by the LLM.

Question	"Who played grandpa joe in charlie and the chocolate factory?" (from ASQA)
DPR + VICUNA 13B	Willy Wonka and the Chocolate Factory is a 1970 children's book by Roald Dahl. It has been adapted into various media, including a 1971 film and a 2005 film. In the 1971 film, Grandpa Joe is played by Wilbur Robinson. In the 2005 film, Grandpa Joe is played by David Morris. – Groundedness: 0.61
SILVER + VICUNA 13B (proposed)	Charlie and the Chocolate Factory is a 2005 musical fantasy comedy film directed by Tim Burton and written by John August, based on the 1964 British novel of the same name by Roald Dahl. The film stars Johnny Depp as Willy Wonka and Freddie Highmore as Charlie Bucket. The storyline follows Charlie, who wins a contest and, along with four other contest winners, is led by Wonka on a tour of his chocolate factory, the most magnificent in the world. In the film, Grandpa Joe is played by David Kelly. In the 2010 musical adaptation of Charlie and the Chocolate Factory, Grandpa Joe is played by Richard Dreyfuss. – Groundedness: 0.94
Question	"Number of episodes in The Sopranos?" (from CONVMIX)
DPR + VICUNA 13B	The Sopranos is an American crime drama television series created by David Chase. The series ran for 73 episodes over six seasons, from January 15, 1999, to June 11, 2007. – Groundedness: 0.23
SILVER + VICUNA 13B (proposed)	The Sopranos is an American crime drama television series created by David Chase. The series portrays the difficulties that fictional New Jersey-based, Italian American mobster Tony Soprano (James Gandolfini) faces as he tries to balance his home life and his criminal organization. The series features Tony's family members, mafia colleagues, and rivals in prominent roles and story arcs, most notably his wife Carmela (Edie Falco) and protégé Christopher Moltisanti (Michael Imperioli). The series ran for 86 episodes over six seasons. – Groundedness: 1.00

Table 3: Anecdotal examples from both datasets (randomly sampled) illustrating the benefits of explicitly retrieving contextual information. Un-grounded tokens (i.e., the ones not appearing in the retrieved passages) are highlighted in orange. The fraction of grounded tokens is greatly improved with our proposed approach in both cases.

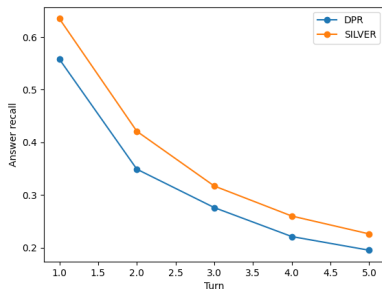


Figure 1: Answer recall per turn on CONVMIX.

3.2 Analysis

Anecdotal examples. Table 3 demonstrates how our approach can improve the groundedness of generated answers: when conditioned on our SILVER retrieval, the LLM can mostly rely on the information in the provided passages for generating LFAs. This can reduce factual hallucinations compared to the DPR-based variant, as illustrated in the second example: as DPR retrieval is insufficient, the LLM hallucinates incorrect contextual information (e.g., that Sopranos started on *January 15, 1999*).

Comparison with state-of-the-art. Our approach achieves state-of-the-art performance on the ASQA dataset, as shown in Table 2.

Answer recall per turn. We plot the average answer recall per turn on CONVMIX in Fig. 1. As expected, the answer recall drops as the conversation drifts away from the initial topic. Recall with our SILVER re-ranker remains consistently higher than for DPR.

We further investigate the effect of the LLM size in Appendix C.

4 Related work

Long-form question answering. With the recent advances of LLMs (Ouyang et al., 2022; Devlin et al., 2019; Zheng et al., 2023) question answering has evolved beyond crisp and direct an-

swers (Yahya et al., 2013; Bast and Haussmann, 2015; Sun et al., 2018; Roy and Anand, 2022) toward supplying more in-depth and comprehensive passage-length responses. There has been extensive research on LFQA recently (Nakano et al., 2021; Amplayo et al., 2023; Jiang et al., 2023; Fan et al., 2019; Su et al., 2022; Wang et al., 2022; Krishna et al., 2021; Gao et al., 2023; Sun et al., 2023), which has mostly built upon retrieval systems optimized for retrieving direct answers.

Weak supervision for training retrieval systems. Obtaining training data for retrieval in QA has been a long-standing challenge (Shen et al., 2023). The most common approach to obtain training samples is to consider all passages matching the direct answer as relevant (Karpukhin et al., 2020; Sachan et al., 2021; Joshi et al., 2017). We extend this approach to the task of LFQA showing how to use both long-form and direct answers for optimizing retrieval toward contextual information.

5 Conclusion

The retrieval of relevant contextual information is often neglected in LFQA, while being an important ingredient for generating and grounding comprehensive long-form answers. We investigate techniques to obtain positive training samples providing such contextual information, and use these samples in training re-ranking models. We show that incorporating these re-rankers improves end-to-end QA performance on a LFQA dataset, significantly boosting the groundedness of generated texts.

Our results indicate that providing LLMs with the right combination of (i) information directly answering the question, and (ii) relevant contextual information, offers the promise of improved retrieval-augmented generation for LFQA.

6 Limitations

Our experimental setup with the ASQA dataset, which has reference long-form answers and short-form answers, allows us to investigate the duality of the LFQA and ConvQA tasks. However, we only evaluated our approach on data that is publicly available. We leave it for future work to run experiments with the approach in the wild.

In this work, we showed improvements on the retrieval side of a RAG pipeline based on Vicuna models of different sizes. RAG pipelines based on other language models architectures might be affected differently by the enhanced retrieval recall provided by our approach, which is not investigated in this paper.

7 Ethical considerations

We did not collect or release any private data or user data in this work. All experiments are based on static datasets.

We make use of LLMs, which are known to generate factually incorrect texts or hallucinations. In this work, we aim to enhance the grounding of long-form answers by explicitly retrieving passages for contextual information. Experiments indicate that with our approach the groundedness of answers can be improved, which is a promising direction of reducing hallucinations in LLMs.

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Algorithm 1 Pseudo-code for SILVER approach

Inputs:

$\mathcal{D}$: Dataset with questions, LFAs and DAs;
 $\mathcal{R}$: pre-trained BERT;
 $\mathcal{M}$: pre-trained causal LLM;
DPR: first-stage retriever;
 k : number of passages in context

Outputs:

$\mathcal{R}_{SFT}$: fine-tuned re-ranker;
 $\mathcal{M}_{SFT}$: fine-tuned causal LLM

```
1: # Identify silver passages
2:  $\mathcal{D}_P \leftarrow \{\}$ ;
3: for all  $(q, LFA, DA) \in \mathcal{D}$  do
4:   # First-stage retrieval (top-100)
5:    $P_{DPR} \leftarrow \text{DPR}(q, 100)$ ;
6:   # Compute matching with LFA
7:   for all  $p \in P_{DPR}$  do
8:      $score(p) \leftarrow match(p, LFA)$ ;
9:   end for
10:  # Compute silver passages  $P^*$ 
11:   $P_{DA}^* \leftarrow \{p | DA \in p\}$ ;
12:   $P_{LFA}^* \leftarrow sort(\{p\}, score)$ ;
13:   $P^* \leftarrow top_k(P_{DA}^*, P_{LFA}^*, score)$ ;
14:   $P^- \leftarrow sample(P - P^*, 50)$ ;
15:  # Add to training data
16:   $\mathcal{D}_P \leftarrow \mathcal{D}_P \cup \{(q, P^*, P^-)\}$ ;
17: end for
18: # Fine-tune re-ranking model
19:  $\mathcal{R}_{SFT} \leftarrow finetune(\mathcal{R}, \mathcal{D}_P)$ ;
20: # Fine-tune LLM
21:  $\mathcal{D}_{LFA} \leftarrow \{\}$ ;
22: for all  $(q, LFA, DA) \in \mathcal{D}$  do
23:    $P_{DPR} \leftarrow \text{DPR}(q, 100)$ ;
24:   # Apply re-ranker
25:    $P_{\mathcal{R}} \leftarrow \mathcal{R}_{SFT}(P_{DPR}, k)$ ;
26:    $\mathcal{D}_{LFA} \leftarrow \mathcal{D}_{LFA} \cup \{(q, P_{\mathcal{R}}, LFA)\}$ ;
27: end for
28:  $\mathcal{M}_{SFT} \leftarrow finetune(\mathcal{M}, \mathcal{D}_{LFA})$ 
```

A Details on matching variants

Lexical matching with LFA. We initially evaluated (i) token recall, (ii) Jaccard similarity between token sets, and (iii) ROUGE-L (Lin, 2004), and found that token recall (as described in Sec. 2) works best.

Semantic similarity with LFA. We compute the semantic similarity between a candidate p and the LFA as follows:

$$match(p, LFA) = Enc(p) \cdot Enc(LFA) \quad (2)$$

where Enc is the text encoder, and $\cdot$ is the dot product.

LLM perplexity of LFA. We compute the LLM perplexity of the target LFA (of length n), given a candidate passage p , as follows:

$$match(p, LFA) = - \sum_{j=1}^n \log P(t_j | C, p, t_1 \dots t_{j-1}) \quad (3)$$

where P is the probability function of the LLM, t_j is the j -th token of the LFA, and C is the same prompt as the one applied during LLM training and inference. C includes a random sample of $k-1$ candidate passages. We observed that adding this random sample substantially improves the performance as it makes the samples closer to the input seen during LLM inference/training, i.e. a context with k passages.

Pseudo-code for SILVER approach. Fig. 1 illustrates the end-to-end workflow of deriving silver passages, fine-tuning the re-ranker, and fine-tuning the LLM on LFQA data.

B Additional details on experiments

Datasets. ASQA has 6,316 ambiguous questions that originate from the Google search log (Kwiatkowski et al., 2019; Min et al., 2020) paired with LFAs written by crowdworkers. Every sample contains a set of alternative question interpretations and a DA corresponding to each of them. An example ambiguous question is “Who played bonnie in gone with the wind?”, with the two interpretations being “Who played bonnie in the gone with the wind film?” and “Who played bonnie in the gone with the wind musical?”. The corresponding DAs are Cammie King and Leilah de Meza in this case. The dataset provides one LFA for each question in the train set, and two LFAs for each question in the dev set, with an average of 64.8 words per LFA (dev set). Since the test set is hidden, we split the train set, using 95% for training and 5% for development, and the original dev set as our test set. We used the official ASQA evaluation code³ to obtain the ROUGE-L, D-F1 and DR metrics. The dataset is licensed under an Apache License 2.0, thereby permitting its use for research purposes.

³<https://github.com/google-research/language/tree/master/language/asqa>

Method	ROUGE-L	D-F1	DR
DPR + VICUNA 7B	40.6	26.0	32.5
DPR + VICUNA 13B	42.5	27.7	34.3
DPR + VICUNA 33B	42.6	27.8	34.3
SILVER + VICUNA 7B	42.3	26.3	33.3
SILVER + VICUNA 13B	43.4	29.0	35.5
SILVER + VICUNA 33B	43.6	29.0	35.6

Table 4: Comparison of end-to-end QA performance with different model sizes of the VICUNA model family.

We use CONVMIX only for the evaluation of the models trained on the ASQA dataset, since CONVMIX does not provide LFAs to train on. We input only the first question from each conversation (3,000 questions, in total) and evaluate whether the generated LFA provides answers to the (first 4) follow-up questions from the conversation. The dataset is licensed under a CC BY 4.0, thereby permitting its use for research purposes.

Implementation details. We implement SILVER re-rankers as cross-encoders based on BERT models (Devlin et al., 2019)⁴ with 110M parameters. The input format is the following: "[CLS] question [SEP] passage_title [SEP] passage_text", and the output is a scalar indicating the relevance of the respective passage. We apply the re-ranker on top-100 DPR results (better performance than for top-1,000). For training the re-ranker, for each question, we randomly sampled 50 negatives (non-silver passages) along with 5 positives (top-5 silver passages) from the top-100 DPR passages. We used AdamW as optimizer with a learning rate of 10^{-5} , batch size of 16, weight decay of 0.01, and warm-up ratio of 0.04. Binary cross-entropy is used as loss function.

Comparison with state-of-the-art. For this comparison, we used the full ASQA train set, same as in related work. Note that the LLM here is trained for one epoch, without optimization on the dev set. As ASQA is a rather small-scale dataset with only 4,353 in the original train set, the additional 5% (compared with our new split) make quite an impact on the LLM performance: the DR metric improves from 35.5 to 36.9.

Computational costs. Fine-tuning the BERT-based re-ranker took 50 minutes on an AWS EC2 P3 instance. The LLM fine-tuning (Vicuna 13B) took 110 minutes on an AWS EC2 P4 instance.

C Additional experiments

Effect of LLM size. We investigate the effect of the LLM size, to verify that our improvements still hold for smaller/larger LLMs, using the 7B and 33B versions of Vicuna. Results are shown in Table 4. In general, the DR metric decreases (SILVER: 33.3; DPR: 32.5) using the 7B version. Further, we found that the effect of scaling up the LLM to 33B parameters is negligible compared to enhancements on the retrieval side, observing very similar results as for the 13B version (DR metric for SILVER: 35.6; DPR: 34.3).

⁴<https://huggingface.co/google-bert/bert-base-uncased>